

- Q.19 Describe the role of computer aided design (CAD) in die design. (CO10)
- Q.20 Explain the significance of maintaining proper alignment in forging dies. (CO9)
- Q.21 What are the typical maintenance practices for casting dies? (CO7)
- Q.22 Outline the steps involved in troubleshooting common defects in forging. (CO8)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Discuss the various types of forging processes and their applications in industry. (CO6)
- Q.24 Explain the complete die casting cycle including all necessary stages and their significance. (CO9)
- Q.25 Analyze the common safety measures that should be implemented in forging and casting operations. (CO7)

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3rd Year / Advance diploma in tool and die making Subject : Tool Design IV (Forging and Casting dies)

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What type of die is used for creating complex shapes in metal? (CO3)
- a) Open die b) Closed die
c) Progressive die d) Punch die
- Q.2 Which forging operation involves shaping the metal into specific profiles using a hammer? (CO2)
- a) Press forging b) Hammer forging
c) Roll forging d) Upsetting
- Q.3 The ejection system in die casting is primarily used to: (CO7)
- a) Shape the mold
b) Inject the molten metal
c) Remove the finished product
d) Heat the mold

- Q.4 In the die casting the term “Clamping force” refer to: (CO2)
- The force required to melt the metal
 - The force used to keep the mold closed
 - The force used to eject the product
 - The force used to shape the die
- Q.5 What is a common material for making die casting dies? (CO1)
- Bronze
 - Steel
 - Cast iron
 - Aluminum
- Q.6 The purpose of using a release agent in casting is to: (CO9)
- Speed up the cooling process
 - Prevent the metal from sticking to the mold
 - Enhance the metal's strength
 - Improve the finish

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The process of shaping metal by forcing it through a die is called _____. (CO2)
- Q.8 The two main categories of casting processes are _____ and _____. (CO4)

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- Q.9 A _____ die allows for the injection of molten metal under pressure. (CO3)
- Q.10 The system responsible for maintaining the temperature in die casting is called the _____ system. (CO6)
- Q.11 Common forging defects include _____ and _____. (CO4)
- Q.12 The term “Flash” in forging refers to _____. (CO7)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What are the primary advantages of using forging over other metal forming processes? (CO1)
- Q.14 Explain the concept of the die layout and its importance in production efficiency. (CO6)
- Q.15 Compare the functions of hot chamber and cold chamber die casting processes. (CO8)
- Q.16 Identify the key design parameters for forging dies. (CO10)
- Q.17 Discuss how temperature affects the forging process. (CO2)
- Q.18 What factors influence the selection of materials for forging and casting dies? (CO6)

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